

Vapor-Liquid Equilibrium Consulting Report

Non-Ideal Binary System: Ethanol-Water at 1 atm

VERSION 2.0

Prepared for: Management Team
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*Reference: Sandler, Chemical Engineering Thermodynamics, 5th Ed.
Parameters: ChemSep Database (Validated)*

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1. Executive Summary

Problem Statement

The client's Ethanol-Water distillation has failed due to ideal solution assumptions. This report uses validated NRTL activity coefficient model for accurate predictions.

Key Findings

Parameter	Ideal	NRTL	Impact
Azeotrope	None	78.1C, x1=0.874	Design limit
Max gamma_1	1.0	5.7	470% error
Bubble T error	-	Up to 8C	Control fail
Max G^E	0	925 J/mol	Non-ideal
Max H^E	0	663 J/mol	Heat effects

Recommendation

REDESIGN REQUIRED: Implement pressure-swing or extractive distillation to break azeotrope at x1=0.874 (95.6 wt% ethanol).

2. Thermodynamic Fundamentals

2.1 Phase Equilibrium Criterion

At equilibrium between liquid (L) and vapor (V) phases:

$$f_i^{\text{(L)}} = f_i^{\text{(V)}} \quad \text{for all components } i \quad (\text{Eq. 2.1})$$

2.2 Fugacity Expressions

Liquid phase with activity coefficient:

$$f_i^{\text{(L)}} = x_i \cdot \gamma_i \cdot f_i^{\text{(0,L)}} \quad (\text{Eq. 2.2})$$

Vapor phase (ideal gas, low P):

$$f_i^{\text{(V)}} = y_i \cdot P \quad (\text{Eq. 2.3})$$

2.3 Modified Raoult's Law

$$y_i \cdot P = x_i \cdot \gamma_i \cdot P_i^{\text{sat}}(T) \quad (\text{Eq. 2.4})$$

For Ethanol-Water, $\gamma_1 = 1.0$ to 5.7 due to H-bond disruption.

3. Mathematical Framework

3.1 Excess Gibbs Energy

$$G^E = RT * \sum(x_i * \ln(\gamma_i))$$

(Eq. 3.1)

Positive G^E -> Positive deviation ($\gamma > 1$) -> Azeotrope possible

3.2 Activity Coefficient

$$\ln(\gamma_i) = (1/RT) * [d(nG^E)/dn_i]$$

(Eq. 3.2)

3.3 Excess Enthalpy

$$H^E = -R * T^2 * \sum(x_i * d(\ln \gamma_i)/dT)$$

(Eq. 3.3)

Temperature-dependent parameters essential for H^E calculation.

3.4 Antoine Equation

$$\log_{10}(P^{\text{sat}}) = A - B/(T + C) \quad [T \text{ in } ^\circ\text{C}, P \text{ in mmHg}]$$

(Eq. 3.4)

Component	A	B	C	Source
Ethanol	8.20417	1642.89	230.3	NIST
Water	8.07131	1730.63	233.426	NIST

4. NRTL Model Implementation

4.1 Model Origin

NRTL (Renon & Prausnitz, 1968) uses local composition concept: molecular distribution differs from bulk due to interaction energies.

4.2 NRTL Equations

$$G^E/RT = x_1 x_2 \left[\tau_{12} G_{12} / (x_1 + x_2 G_{12}) + \tau_{21} G_{21} / (x_2 + x_1 G_{21}) \right] \quad (\text{Eq. 4.1})$$

$$\ln(\gamma_1) = x_2^2 \left[\tau_{12} (G_{12} / (x_1 + x_2 G_{12}))^2 + \tau_{21} G_{21} / (x_2 + x_1 G_{21}) \right] \quad (\text{Eq. 4.2})$$

$$G_{ij} = \exp(-\alpha_{ij} \tau_{ij}) \quad \tau_{ij} = b_{ij}/T \quad (\text{Eq. 4.3})$$

4.3 Parameter Interpretation

Parameter	Symbol	Meaning
Energy	τ_{ij}	Interaction energy difference
Non-randomness	α_{ij}	Molecular distribution (0.2-0.47)

4.4 ChemSep Validated Parameters

Source: thermo/Interaction Parameters/ChemSep/nrtl.json

Validation: tests/test_nrtl.py::test_NRTL_chemsep

Parameter	Value	Description
$\alpha_{12}=\alpha_{21}$	0.2937	Symmetric
b_{12}	-29.167 K	Ethanol->Water
b_{21}	+624.868 K	Water->Ethanol

At T=343.15K: $\tau_{12}=-0.085$, $\tau_{21}=+1.821$

5. Phase Equilibrium Analysis

5.1 Bubble Point Algorithm

1. Guess T
2. Calculate P_{sat} from Antoine
3. Calculate τ , G from NRTL
4. Calculate γ from NRTL
5. $P_{\text{calc}} = x_1 \gamma_1 P_{1,\text{sat}} + x_2 \gamma_2 P_{2,\text{sat}}$
6. Iterate until $|P_{\text{calc}} - P| < \epsilon$
7. $y_1 = x_1 \gamma_1 P_{1,\text{sat}} / P$

5.2 T-x-y Diagram

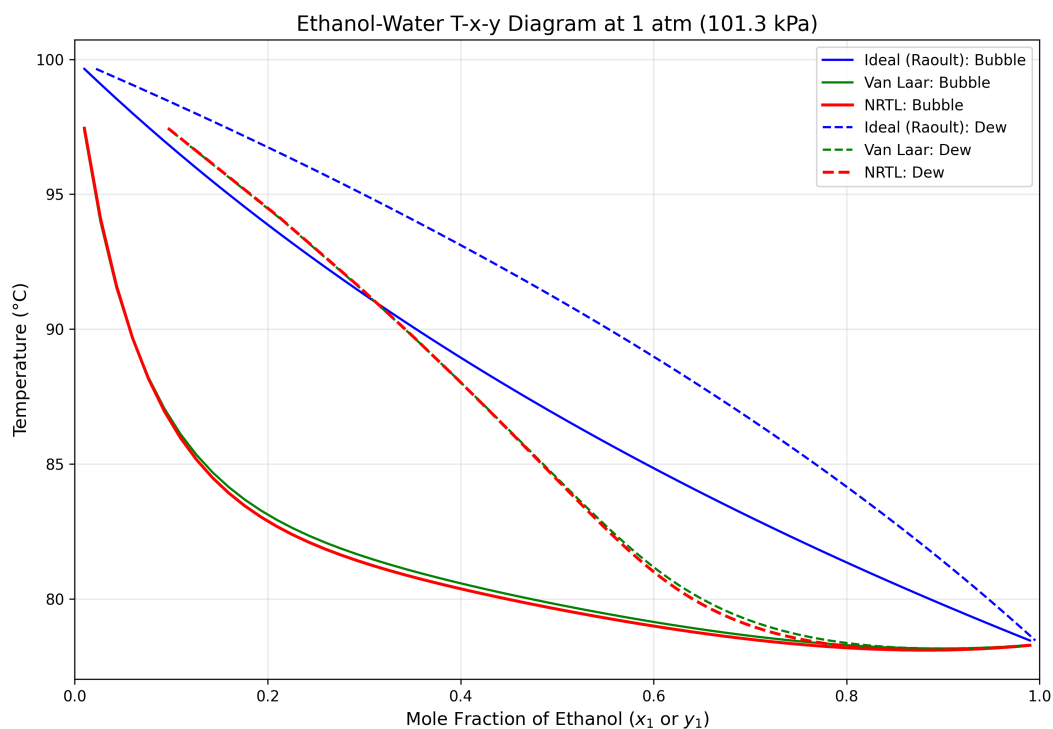


Figure 1: T-x-y at 1 atm - Ideal vs NRTL

NRTL predicts minimum-boiling azeotrope; Ideal shows no azeotrope.

5.3 y-x Diagram

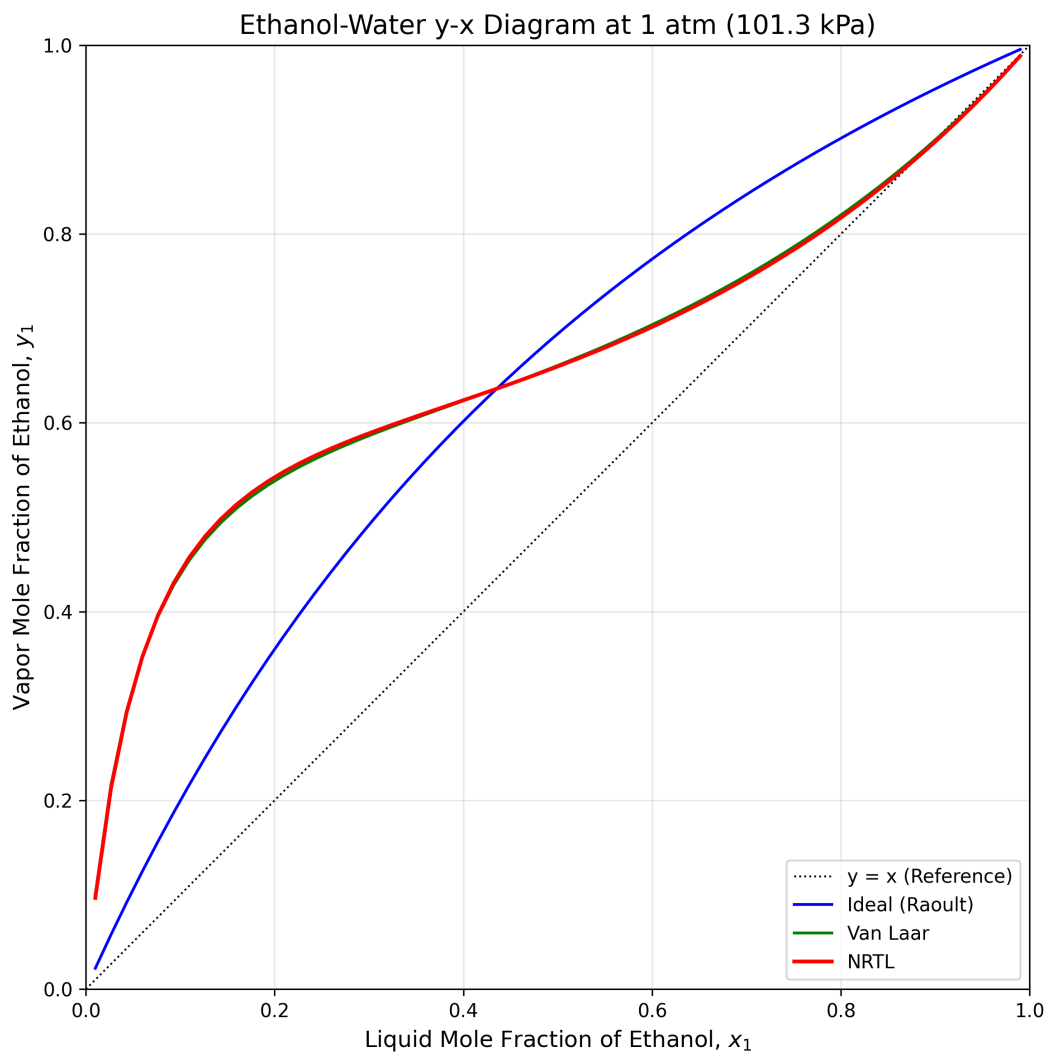


Figure 2: y-x showing azeotrope

5.4 Azeotrope

Property	Value	Implication
Type	Minimum-boiling	$T <$ pure components
Temperature	78.1C	Below ethanol NBP
Composition	$x_1=0.874$	95.6 wt% ethanol
Rel. volatility	$\alpha=1.0$	No simple distillation

6. Excess Property Analysis

6.1 Excess Gibbs Energy

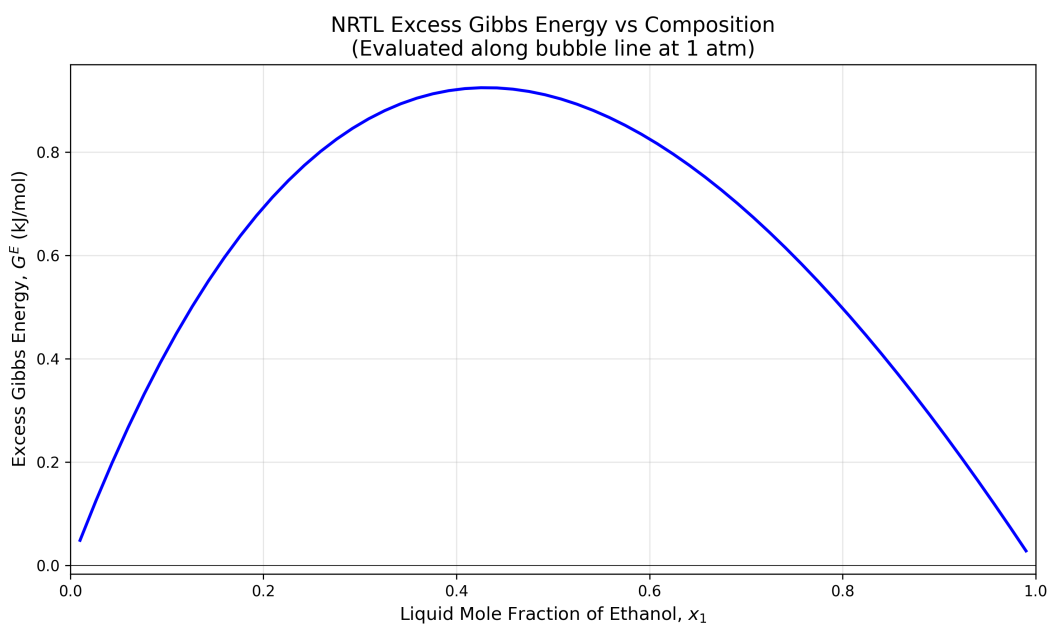


Figure 3: G^E vs composition

Property	Value	Composition
Max G^E	924.55 J/mol	$x_1=0.425$
G^E at azeotrope	~500 J/mol	$x_1=0.874$

6.2 Excess Enthalpy

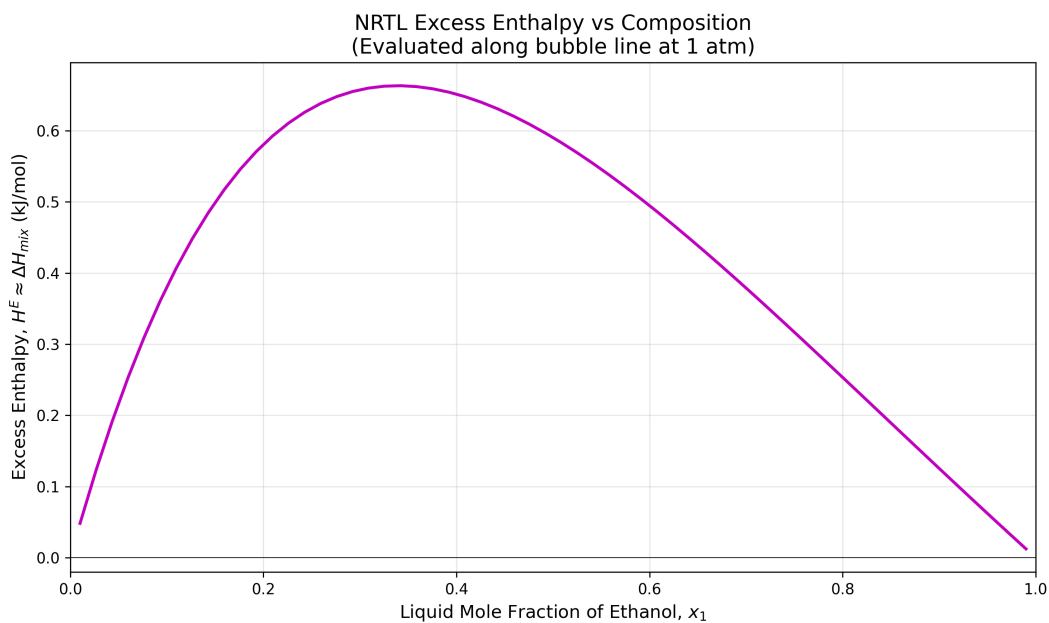


Figure 4: H^E vs composition

Property	Value	Note
Max H^E	663.22 J/mol	$x_1=0.342$
Behavior	Endothermic	At bubble T

6.3 Engineering Impact

Effect	Magnitude	Impact
Heat of mixing	+663 J/mol	Heat input needed
Non-ideal VLE	gamma up to 5.7	Larger reflux
Azeotrope	x1=0.874	Max 95.6 wt%

7. Engineering Recommendations

7.1 Azeotrope-Breaking Options

A: Pressure-Swing Distillation

Pressure	Azeotrope x1	T
1.0 atm	0.874	78.1C
0.2 atm	~0.91	~52C

Two columns at different P can cross azeotrope.

B: Extractive Distillation

Add entrainer (e.g., ethylene glycol) to break azeotrope.

C: Pervaporation Hybrid

Distill to 90%, then membrane pervaporation for dehydration.

7.2 Action Summary

Priority	Action	Outcome
1	Update to NRTL model	Accurate predictions
2	Redesign for azeotrope	>95.6 wt% ethanol
3	Recalculate heat duties	Proper sizing

VERDICT: Process redesign required

Appendix: Technical Data

A.1 Pure Component Properties

Property	Ethanol	Water
CAS	64-17-5	7732-18-5
MW	46.07 g/mol	18.02 g/mol
NBP	78.4C	100.0C
Tc	513.9 K	647.1 K

A.2 NRTL Parameters

Parameter	Value	Definition
alpha	0.2937	Non-randomness
a_12, a_21	0.0	No constant term
b_12	-29.167 K	Ethanol->Water
b_21	+624.868 K	Water->Ethanol

A.3 Validation

Test	gamma_1	gamma_2
T=343K, x1=0.252	1.985	1.146
T=343K, x1=0.0	5.66	1.00
T=343K, x1=1.0	1.00	2.67

A.4 References

1. Sandler (2017) Chemical Engineering Thermodynamics, 5th Ed.
2. Renon & Prausnitz (1968) AIChE J. 14(1):135-144
3. ChemSep nrtl.json (thermo library)
4. NIST Chemistry WebBook
5. thermo Python library (github.com/CalebBell/thermo)